

Complex Engineering Project (CEP)

CSE 345 : Information System Design & Software Engineering

General Instructions: This is a Complex Engineering Project (CEP). Originality and independent thinking are mandatory the scenario is deliberately open-ended so that no two students should produce identical designs. All submitted work must demonstrate individual analysis, judgment, and creativity. Plagiarism in any form will result in a zero mark for the entire submission. All diagrams must be produced using proper CASE tools (draw.io, StarUML, PlantUML, Lucidchart, or equivalent). Hand-drawn diagrams are not accepted.

Course Outcomes & Learning Taxonomy

COs	Descriptions	POs	Domain/ level of learning taxonomy
CO1	Discuss the software development life cycle and evolve methodologies for the design, analysis, planning and development of an information system.	PO2	Cognitive / L3, Psychomotor/ L3 Affective / L3
CO2	Use different project management, information gathering and requirements engineering principles to a S/W project.	PO4	Psychomotor/ L2 Affective / L2
CO3	Construct, analyze, combine and document UML and design patterns to work together in software design.	PO3	Cognitive / L3, Affective / L3
CO4	Implement software testing and estimation techniques professionally for developing various categories of softwares subject to budget, schedule and platform constraints.	PO1	Cognitive / L3, Affective / L3

The Problem

Digitization of an Inland Waterway Freight Network Under Multi-Stakeholder, Multi-Constraint Conditions

Bangladesh's inland waterway freight network serving over 700 rivers and connecting 11 major river-port terminals across 7 districts has long depended on manual, paper-based processes for vessel scheduling, cargo documentation, regulatory clearance, and workforce coordination. Following Cyclone Mahaan in October 2025, which rendered 38 road bridges structurally unsafe for heavy cargo, the inland network became the primary logistics corridor for approximately 60% of the country's inter-district freight, exposing the full scale of these systemic failures.

A newly established public-private consortium, NadiPath Multimodal Logistics Ltd., has been mandated by the government and co-financed by the World Bank (USD 120 million, conditional on delivery of a functional, ISO-auditable digital platform) to address this crisis. The consortium brings together the Bangladesh Inland Water Transport Authority (BIWTA), three private shipping conglomerates, the Department of Disaster Management, and a sovereign investment fund.

The operational problems NadiPath faces are not merely technical. They are deeply multidisciplinary in nature spanning engineering, economics, law, human behavior, and public administration.

The challenge is further complicated by a set of hard constraints that any proposed approach must contend with: four terminals operate in areas with intermittent network connectivity; a 2009-era vessel registration system cannot be replaced within the project timeline; multiple regulatory bodies (NBR, BIWTA, BSTI) operate disconnected legacy platforms; stakeholder groups have conflicting interests (private operators resist data sharing with competitors; small informal traders fear regulatory exposure from digitization; dock workers have previously resisted and derailed two earlier BIWTA digitization attempts); and the entire Phase 1 delivery must occur within 14 months on a budget of USD 8.5 million.

NadiPath's leadership requires a rigorous, evidence-based software engineering approach that identifies and structures the operational problems, proposes and justifies a viable design and implementation strategy, and demonstrates awareness of the technical, economic, human, and regulatory dimensions of the challenge. There is no single correct solution , students are expected to critically analyze the problem, evaluate trade-offs, and defend their chosen approach.

Assignment Tasks

The following four tasks must each be addressed as clearly separated, titled sections in your report. All design decisions must be traceable to the specific details of the NadiPath scenario; generic or template-style answers will not receive full marks.

Task Description	CO	PO / WK / WP	Marks
1. Identify and structure the operational, technical, and organizational challenges embedded in the NadiPath scenario into a set of clearly defined business and software requirements. Your analysis should distinguish between different categories of requirements, prioritize them based on criticality to the network's core failures, and reflect the perspectives of the multiple stakeholder groups involved including those with conflicting interests. Justify your prioritization framework and explain what trade-offs emerge	CO1	PO1, PO2 WK3, WP1, WP6	5

when requirements from different stakeholders cannot all be satisfied simultaneously.			
2. Develop a structured software design for the proposed system using appropriate UML modelling techniques. Your design should capture the system's structural organization, dynamic behavior, and key interactions between actors and system components. Select the diagram types that best communicate the complexity of the problem you must include at minimum four distinct diagram types drawn from the UML standard. For each diagram, provide a written analytical commentary that explains the design decisions made, the alternatives considered, and the specific aspects of the NadiPath scenario that informed those decisions.	CO2	PO3 WK5, WP1, WP2	10
3. Select and justify appropriate tools, platforms, technologies, and development methodologies to implement the proposed system under the constraints described in the scenario. Your justification must go beyond feature comparisons, it should address how each choice responds to specific constraints (budgetary, connectivity, legacy integration, regulatory, and human factors) and what risks are introduced or mitigated by that choice. Where genuine uncertainty or trade-offs exist, these should be explicitly acknowledged rather than hidden.	CO3	PO4, WK6, WP1, WP2	5
Total			20

Assessment Rubric

Task	Excellent (90–100%)	Satisfactory (60–89%)	Needs Improvement (<60%)
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Submission Guidelines

- ❖ Submit a single PDF document. Include a cover page (student name, ID, section, date), a table of contents with page numbers, and clearly numbered sections for each task.
- ❖ Gantt Chart must be produced.
- ❖ All figures, tables, and diagrams must be numbered and captioned.
- ❖ Maximum report length: 20 pages.
- ❖ Submission will be in form of typed.
- ❖ Late submissions will be penalized 10% per day.